



MODULES IN ANGULAR



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✓ What is Modules?

📖 In Angular, a module is a container that groups related components, directives, pipes, and services.

Each module is defined using the @NgModule decorator.

🧱 Modules help in:

- Organizing code
- Lazy loading features
- Reusability & separation of concerns

✓ Anatomy of a Module

An Angular module (AppModule, for example) includes:

```
@NgModule({  
  declarations: [AppComponent],  
  imports: [BrowserModule],  
  providers: [],  
  bootstrap: [AppComponent]  
})  
export class AppModule {}
```

📌 Key properties:

- **declarations** → Components, directives, pipes
- **imports** → Other modules
- **providers** → Services
- **bootstrap** → Root component to launch

✓ AppModule – The Root Module

⚙️ Every Angular app has at least one root module: AppModule.

Responsible for:

- Bootstrapping the app
- Bringing in shared and feature modules



```
platformBrowserDynamic().bootstrapModule(AppModule);
```

It's the entry point of your Angular app.

✓ Feature Modules

Feature modules organize code into functional areas:

Examples:

- UserModule
- AdminModule
- ProductModule

```
● ● ●  
  
@NgModule({  
  declarations: [UserProfileComponent],  
  imports: [CommonModule, FormsModule]  
})  
export class UserModule {}
```

✓ Shared Module

Use a Shared Module to avoid code duplication:

```
@NgModule({  
  declarations: [CommonComponent],  
  exports: [CommonComponent],  
  imports: [CommonModule]  
})  
export class SharedModule {}
```

Import SharedModule into multiple feature modules to share components, pipes, or directives.

✓ Lazy Loading Modules

Boost performance by loading modules only when needed:

```
const routes: Routes = [  
  { path: 'products', loadChildren: () =>  
    import('./product/product.module').then(m => m.ProductModule) }  
];
```

🎯 Improves initial load time

📦 Keeps app size optimized

✓ Core Module (Best Practice)

Create a CoreModule for singleton services and global components:

```
    @NgModule({  
      providers: [AuthService]  
    })  
    export class CoreModule {}
```

🎯 Imported once in AppModule only

📦 Central place for app-wide services

Key Takeaways

- **Angular modules organize components & features logically**
- **Use AppModule to bootstrap**
- **Create Feature Modules to scale**
- **Use Shared/Core Modules for reusability and singleton services**
- **Apply lazy loading for performance**



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